

Machine Learning for Natural Language Understanding

Prof. Achim Rettinger

Language Models

**Cats Eyes
Removed**

Lecture 5 – Language Models 1

Sources Lecture and Exercise

- Language Models and RNNs
 - <https://youtu.be/PLryWeHPcBs?t=3000> (Intro and RNN basics)
 - <https://youtu.be/OLixFSa7yts> (starts with training RNNs)
 - <https://youtu.be/OLixFSa7yts?t=1091> (Generating text with RNNs)
 - <https://youtu.be/OLixFSa7yts?t=1459> (Evaluating Language Models (Perplexity))
 - <https://youtu.be/OLixFSa7yts?t=1772> (Applications of LMs)
 - <https://youtu.be/OLixFSa7yts?t=2218> (Issues (Vanishing Gradient))
 - <https://youtu.be/OLixFSa7yts?t=2983> (LSTMs)
 - <https://youtu.be/OLixFSa7yts?t=4596> (Bi-LSTMs and Deep LSTMs)
- Attention in RNNs
 - <https://youtu.be/wzfWHP6SXxY?t=4367> (Seq2Seq tasks)

Flipped Classroom Questions

- What is a “neural language model”? What is the advantage over n-grams.
- What is the difference between contextualized word embeddings vs static word embeddings (like e.g., from word2vec)?
- How do RNNs overcome the limited window size?
- How many gates are in an LSTM?

“The future of language is about the development of language models...”¹

1

Computerlinguistik | Universität Trier | cl.cldh.uni-trier.de